



Gisma
University
of Applied
Sciences

Gisma University of Applied Sciences
School of Computer Science

Module Handbook

F107 Foundations of Computer Science



Module Summary

Code:	F107
Name:	Foundations of Computer Science
Level:	Foundation
Credits:	5 ECTS
Workload:	50 Hours Directed Study + 75 Hours Independent Study
Delievery Mode:	Hyflex
Prerequisites:	None
Corequisites:	None
Leader:	Prof. Dr. Mohammad Mahdavi

Description

The ability to understand and work with computer systems has been recognized as one of the most important skills of this century. In this module, students will learn the foundation of computer science.

Objectives

Specific objectives include

- To explore the core concepts of computer science.
- To work with computer systems in action.

Learning Outcomes

On successful completion of the module, students will be able to

1. Understand the business contexts that can benefit from computer systems.
2. Analyse core computer science concepts, including its hardware and software components.
3. Work with computer systems in various scenarios.

Teaching Methods

Students will engage in learning through a series of directed study activities, lectures, and self-study. The teaching sessions will be divided into two parts. The first part belongs to an interactive lecture, where students learn about theoretical aspects of the presented topic. The second part will be an interactive lab, where students will implement the presented concepts themselves under the guidance of the tutor. Students will be also directed to online materials for further study. Note that teaching sessions will not be recorded by default, but the tutor may decide to record certain parts of lectures from time to time.

Content Outline

Week	Topics	Learning Outcomes	References	Activities
1	Introduction to Computer Science; Number Systems	All	Ch. 1 and 2 [1]	Interactive Lecture; Practical Exercises
2	Data Operations	All	Ch. 3 and 4 [1]	Interactive Lecture; Practical Exercises
3	Computer Organization	All	Ch. 5 [1]	Interactive Lecture; Practical Exercises; Summative Assessment
4	Operating Systems	All	Ch. 7 [1]	Interactive Lecture; Practical Exercises
5	Computer Networks	All	Ch. 6 [1]	Interactive Lecture; Practical Exercises; Summative Assessment
6	Programming Languages; Software Engineering	All	Ch. 9 and 10 [1]	Interactive Lecture; Practical Exercises
7	Data Structures and Algorithms	All	Ch. 8, 11, and 12 [1]	Interactive Lecture; Practical Exercises; Summative Assessment
8	Databases	All	Ch. 13 and 14 [1]	Interactive Lecture; Practical Exercises
9	Artificial Intelligence	All	Ch. 18 [1]	Interactive Lecture; Practical Exercises; Summative Assessment

References

- [1] Behrouz Forouzan. *Foundations of computer science*. Cengage Learning EMEA, 2017.